Valence-Selective Transfer in Frozen Speech-Emotion Embeddings for Multilingual Voice-Based Stress Detection

Abstract—Voice-based stress detectors increasingly reuse frozen speech-emotion (SER) embeddings, and the checkpoint is usually chosen by how well it agrees with acted emotional speech in-domain. We find that this criterion reverses on real, spontaneous stress: over six fusion checkpoints, stronger in-domain arousal agreement goes with weaker real-voice accuracy (Spearman $\rho = -0.60$), and the checkpoint the convention would reject (0.35 in-domain) is the one that deploys best (91.7% leave-one-out). The reason is a split in what the frozen embeddings preserve: they retain the valence of real stress but lose its arousal. On held-out real clips, stressed speech reads as correctly negative in valence 82–100% of the time yet correctly high in arousal only 12–53% of the time. The same gap reappears on a large independent corpus (IEMOCAP, n = 2,132), where predicted valence tracks the labels far more closely than arousal (r = 0.688 vs. 0.485; bootstrap intervals on the per-clip difference clear zero for every out-of-domain checkpoint). A matched-size probe shows this is an effect of emotion pretraining rather than capacity: at equal width emotion2vec encodes valence much better than WavLM (CCC 0.660 vs. 0.387), while WavLM leans toward arousal instead—so the valence advantage is encoder-specific. Following the diagnosis, a valence-primary scoring rule widens the stressed/calm gap over the usual arousal-based rule for every checkpoint (Cohen's d = 1.71). We ship the result as a bilingual detector for English and Sinhala that hands arousal magnitude to a physiological module by design, and we state plainly where a frozen encoder fails out of distribution.

Index Terms—voice stress detection, speech emotion recognition, valence and arousal, frozen embeddings, emotion2vec, prosody, multilingual, low-resource, affective computing.

I. Introduction

Speech carries reliable traces of psychological stress, and reading stress from the voice is appealing because it is passive, cheap, and needs nothing more than a microphone. In well-being systems that adapt as a session unfolds—say a meditation environment that responds to how stressed a student or desk-worker is right now—the voice is often the only affective signal on hand without extra wearables. The real question is therefore not whether stress leaves a vocal trace, but whether it can be read reliably from a few seconds of real, spontaneous speech by an unseen speaker, in more than one language, and without gathering a large labelled in-domain corpus first.

The standard fix for the data problem is a frozen SER encoder. Self-supervised and emotion-pretrained models—wav2vec 2.0 [1], WavLM [2], emotion2vec [3]—provide a fixed high-dimensional representation that a small head maps to affect, so only a few hundred thousand parameters need training instead of a 300M-parameter backbone. Since affect is conventionally placed in the two-dimensional valence–arousal plane of the circumplex model [4], these heads are trained as regressors on valence and arousal and scored with the concordance correlation coefficient (CCC) [5]. The resulting habit is rarely stated but widely followed: pick the checkpoint with the highest in-domain valence/arousal CCC and expect it to deploy best. It is a convenient habit, because in-domain CCC is cheap to measure on the acted benchmarks these models are trained on—IEMOCAP [6], MELD [7], and the like.

The habit assumes something that has not been checked: that matching acted emotional speech predicts accuracy on real stress. Acted corpora are voiced by performers told to portray an emotion, and they lean heavily toward the loud, high-arousal corner of the space [8]. Real stress—in students and desk-workers especially—often shows up instead as a quiet, flattened "freeze": negative in valence but low in arousal. If what a frozen encoder carries over from acted data is not what separates real stress, then ranking checkpoints by in-domain CCC may be useless—or worse, backwards. As far as we know this has not been tested directly for real-voice deployment: a distribution shift can drag down overall accuracy without revealing which affective axis actually survives it.

We test it directly. With six fusion checkpoints spanning the acted-to-natural range, a held-out set of real stressed and calm clips, and a large independent corpus for replication, we ask whether in-domain accuracy predicts real-voice accuracy—and if not, why. The result is a clean negative with a mechanism behind it. First, the criterion does not just fail to predict deployment—it inverts it: across the six checkpoints, higher in-domain arousal CCC comes with lower real-voice accuracy (Spearman $\rho = -0.60$), so the checkpoint ranked worst by convention (0.35 in-domain) deploys best (91.7% leave-one-out). Second, the inversion has a definite cause: frozen embeddings preserve the valence of real stress but not its arousal. On real stressed clips valence is correctly negative 82–100% of the time while arousal is correctly high only 12–53% of the time; the same asymmetry replicates on IEMOCAP (n = 2,132), where predicted valence tracks the labels much better than arousal (r = 0.688 vs. 0.485) and the bootstrap interval on the difference clears zero for every out-of-domain checkpoint. Fig. 1 sketches the picture: the valence axis that tells real stress from calm survives, while the arousal axis on which real "freeze" stress differs from acted stress collapses.

The diagnosis leads straight to an action. If valence is what the embeddings recover for real stress, then stress should be read from valence, not from arousal. A valence-primary rule widens the stressed/calm gap over the arousal-based rule on all six checkpoints, with a large effect (Cohen's $d = 1.71$). With a matched-size probe we further show the valence edge comes from emotion pretraining, not model size: at equal width emotion2vec carries valence far better than WavLM (CCC 0.660 vs. 0.387), whereas WavLM tilts the other way, toward arousal. The split is thus encoder-specific—which is exactly why valence-primary scoring on an emotion-pretrained encoder is a design choice, not a universal rule.

Because our target is bilingual, we test in English and Sinhala. English, well covered by the encoder's pretraining, separates strongly; Sinhala, a low-resource language absent from it [9], marks the honest limit of a frozen encoder out of distribution—one that more in-language data does not lift. The deployed system handles this by design: arousal magnitude, which the voice does not supply dependably, is handed to a separate physiological (heart-rate-variability) module, and the voice contributes only the valence-driven estimate it can support. The paper is therefore diagnosis and design together: why a common selection habit fails on real stress, and what to do instead.

Contributions.

- We show that ranking frozen SER models by in-domain acted CCC is anti-correlated with real-voice stress accuracy (Spearman $\rho = -0.60$), reversing a common selection habit.
- We trace the cause to a valence–arousal split—frozen emotion embeddings carry the valence of real stress but not its arousal—and replicate it at scale on IEMOCAP ($n = 2{,}132$) with bootstrap significance.
- We show the split is encoder-specific to emotion pretraining with a matched-size probe (emotion2vec vs. WavLM), not an artefact of size or architecture.
- We derive and validate a valence-primary scoring rule that follows from the diagnosis and widens stressed/calm separation on every checkpoint (Cohen's $d = 1.71$), deployed as a bilingual (English/Sinhala) detector that defers arousal to a physiological module.
- We report the negative and limiting results plainly—the small-sample and out-of-distribution Sinhala limits, including a controlled case where adding in-language data hurt accuracy—as findings in their own right.

II. Related Work

Frozen speech representations. Self-supervised encoders learn general-purpose features from unlabelled

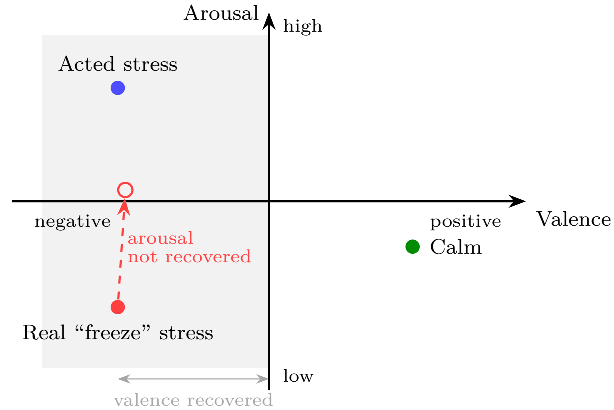

Fig. 1. The valence–arousal split (schematic). Both acted and real "freeze" stress are negative in valence, but acted stress sits high in arousal while real freeze stress sits low. Frozen embeddings keep the valence (horizontal) gap between real stress and calm, yet pull its arousal (vertical) gap toward neutral (dashed arrow): valence survives, arousal does not. This is what motivates the valence-primary rule in the deployed detector.

audio and are now the default front-end for speech tasks. wav2vec 2.0 [1] and WavLM [2] optimise masked-prediction or contrastive objectives on large collections and are commonly used with a frozen backbone under a lightweight probe. emotion2vec [3] adapts this recipe to affect, pretraining on emotional speech so its features track emotion rather than phonetic content, and recent transfer systems reuse such embeddings—at times alongside speaker vectors—as fixed features for classification [10]. We use a frozen emotion2vec encoder in this same probe-on-top setting, but ask a question the benchmarks do not: when deployment departs sharply from acted training data, which affective axis a frozen encoder still supports.

Dimensional emotion and its evaluation. The valence–arousal circumplex [4] underlies most affect models, and continuous recognition is typically scored with CCC [5] against valence and arousal on acted corpora such as IEMOCAP [6] and MELD [7]. That acted and naturalistic emotion differ in both distribution and recoverability [8] is well established and motivates our question; but this line of work tends to treat the two axes symmetrically when transfer is poor, whereas we show the failure is asymmetric and axis-specific.

Voice-based stress in the wild. Reading affect and stress from real speech has a long history, often with prosody-centred feature sets such as GeMAPS [11]. Nearest to us, EMOVOME [12] recognises emotion from spontaneous voice messages by fusing pretrained embeddings with eGeMAPS features, and reports that spontaneous speech trails acted corpora while matching elicited data. That work, like ours, establishes that real speech is not acted speech; our addition is to explain the failure—valence transfers, arousal does not—and to turn it into a concrete selection-and-scoring rule rather than an accuracy figure. More broadly, such systems read stress as a high-arousal state, an assumption our results complicate for internalised stress, where the arousal cue is faint and valence is the dependable signal.

Multilingual and low-resource SER. Multilingual SER has been approached with transformers trained under augmentation and cross-corpus fusion [13], and cross-lingual pretraining helps languages seen in training. Genuinely under-resourced languages—Sinhala among them [9]—still fall out of distribution for encoders trained mostly on high-resource speech. Against the augmentation-helps narrative, we report a controlled case where adding in-language Sinhala data lowered held-out accuracy, and so treat Sinhala not as a solved multilingual case but as a controlled probe of frozen-encoder behaviour out of distribution.

Position. Our contribution is diagnostic. We propose no new encoder or benchmark; we show a common selection convention is not merely uninformative but inverted on real stress, explain the inversion through a valence–arousal split we replicate at scale, and derive the scoring rule that follows from it.

III. Method

A. Data

The real-voice set is $n = 24$ held-out clips (17 stressed, 7 calm) from consenting speakers—13 English and 11 Sinhala from roughly three speakers. Every split here is speaker-independent. For large-n replication we use 2,132 held-out IEMOCAP [6] utterances, each with continuous valence and arousal labels. The six fusion checkpoints differ only in training corpus and so span the acted-to-natural range: Fusion-Acted (acted emotion), Fusion-V2, Fusion-Combined, Fusion-Augmented, Fusion-IEMOCAP (in-domain on IEMOCAP), and Fusion-MELD (the shipped model, trained on the naturalistic conversational MELD corpus [7]).

B. Model

Fig. 2 gives the architecture. Audio passes through a frozen emotion2vec-plus-large encoder [3] (∼300M parameters, feature extractor only) yielding a 1024-d embedding $\mathbf{e}_0$, and through a prosodic front-end computing 21 interpretable features $\mathbf{p}_0$ (pitch statistics, jitter, shimmer, harmonics-to-noise ratio, RMS energy, zero-crossing rate, spectral centroid/rolloff/flatness, speech rate, pause ratio, voiced fraction, duration) in the spirit of GeMAPS [11]. A gated fusion head—the only trainable part (∼1–2M parameters)—mixes the two streams and regresses valence and arousal:

$$\begin{gathered}\mathbf{e} = \mathrm{ReLU}(\mathrm{LN}(\mathbf{W}_e\mathbf{e}_0)), \quad \mathbf{p} = \mathrm{ReLU}(\mathrm{LN}(\mathbf{W}_p\mathbf{p}_0)),\\ \mathbf{g} = \sigma(\mathbf{W}_g[\mathbf{e};\mathbf{p}]), \quad \mathbf{f} = \mathbf{g}\odot\mathbf{e} + (1-\mathbf{g})\odot\mathbf{p}, \\ (\hat{v},\hat{a}) = \tanh(\mathrm{head}(\mathbf{f})).\end{gathered} \tag{1}$$

Here $\mathbf{W}_e : 1024 \to 128$ and $\mathbf{W}_p : 21 \to 128$, the gate $\mathbf{g} \in (0,1)^{128}$ mixes streams elementwise, and the head maps $128 \to 64 \to \mathrm{ReLU} \to \mathrm{Dropout}(0.3) \to 2$ to $\hat{v}, \hat{a} \in (-1, 1)$. Only the head, fusion, and prosody projections train—the encoder stays frozen—under a concordance loss with a small MSE term, $\mathcal{L} = 1 - \frac{1}{2}(\mathrm{CCC}_v + \mathrm{CCC}_a) + 0.2\,\mathrm{MSE}$.

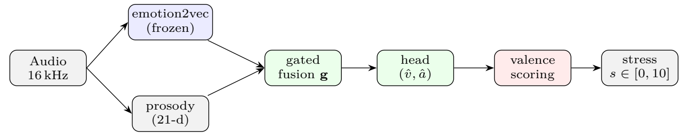

Fig. 2. System architecture. The 300M-parameter emotion2vec encoder is frozen and used only as a feature extractor; the prosodic front-end and gated fusion head (blue = frozen, green = trained, ∼1–2M parameters) are the only trained parts. The head predicts valence and arousal, but only valence drives the deployed score.

C. Scoring

The legacy rule reads stress off arousal; our valence-primary rule reads it off valence, $s = \mathrm{round}(10\max(0, -\hat{v}), 2) \in [0, 10]$ with confidence $c = \min(1, |\hat{v}|)$. A clip is flagged stressed when $s > 2.0$, a threshold set by leave-one-out validation. Arousal magnitude is deliberately not used as a stress magnitude; the system hands it to a separate heart-rate-variability module.

D. Preprocessing

Training and inference share one pipeline: resample to 16 kHz mono, a 70 Hz high-pass filter, silence trimming, and loudness normalisation to RMS 0.05. A lightweight quality gate drops clips that are too quiet, clipped, or short on detected voice before scoring.

E. Statistical protocol

Given the small real-voice set, we keep effect estimation and inference apart. On the 24 clips and six checkpoints we report effect sizes (Cohen's d [14]), Spearman correlations, and bootstrap 95% intervals (10,000 resamples) [15], and we deliberately do not treat the underpowered p-values as significant. Inference is carried instead by the IEMOCAP replication ($n = 2{,}132$). Deployment accuracy uses leave-one-out (LOO) thresholding, which removes the bias of tuning a threshold on the same clips it is scored on.

IV. Results

A. The inversion

Table I and Fig. 3 plot in-domain acted arousal CCC (the selection criterion) against real-voice accuracy under legacy scoring. The trend is monotone and negative: as in-domain agreement rises, real-voice accuracy falls (Spearman $\rho = -0.60$). The weakest in-domain checkpoint, Fusion-MELD (0.35), is the strongest deployed (0.917); the strongest in-domain, Fusion-Acted (0.81), is the weakest deployed (0.375). With five distinct in-domain points the correlation is directional, not significant—we claim no significance here—but the ordering is unambiguous and the

TABLE I
The inversion: in-domain acted arousal CCC versus real-voice accuracy (legacy scoring, $n = 24$). Higher in-domain score → lower deployed accuracy.

Checkpoint	In-domain acted CCC	Real-voice acc.
Fusion-Acted	0.81	0.375
Fusion-V2	0.79	0.750
Fusion-Combined	0.77	0.625
Fusion-IEMOCAP	0.65	0.583
Fusion-MELD (shipped)	0.35	0.917
Spearman $\rho = -0.60$ (directional; $n = 5$)		

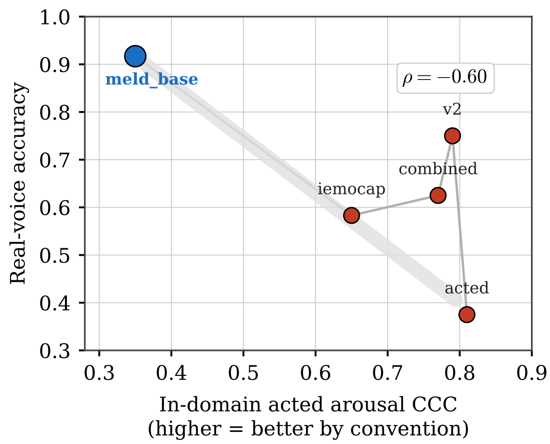

Fig. 3. The inversion, visualised. Each point is a checkpoint; the trend falls as in-domain acted CCC rises. The checkpoint convention would discard (top-left, low in-domain) deploys best.

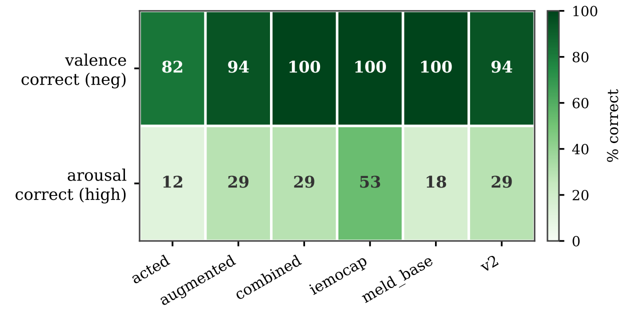

Fig. 4. Valence transfers, arousal does not. On real stressed clips, valence is correctly negative 82–100% of the time; arousal is correctly high only 12–53% of the time, across all six checkpoints.

mechanism below explains it. Under the shipped valence-primary scoring the inversion eases to $\rho = -0.50$, since the fix raises every checkpoint's accuracy: the convention is harmful under legacy scoring and merely uninformative once scoring is corrected.

B. The split on real clips

Separating the two axes exposes the cause. Fig. 4 shows, on real stressed clips, how often each checkpoint puts valence on the correct (negative) side versus arousal on the correct (high) side. Valence is recovered almost perfectly—82 to 100%, hitting 100% for the shipped Fusion-MELD—while arousal is recovered only 12 to 53% of the time. Real stress reliably reads as negative in valence but fails to read as high in arousal, exactly the pattern of Fig. 1: the quiet "freeze" of genuine stress lands in the low-arousal region that acted training never stresses.

C. Replication on IEMOCAP

Twenty-four clips cannot support a significance claim, so we replicate the split on the 2,132-utterance IEMOCAP test set, whose valence and arousal labels let us correlate predictions with ground truth directly. Table II gives predicted-versus-true Pearson r. For every checkpoint trained outside IEMOCAP, valence correlates well above arousal, and the bootstrap interval on the per-utterance difference $r_v - r_a$ clears zero in every case—the split is significant at scale. Two honest nuances follow. First, arousal is degraded, not collapsed, on IEMOCAP: its expressive acted delivery still leaves an arousal signal (r_a up to 0.62), whereas the full collapse shows up on quiet real "freeze" stress—so the severity of the split scales with how internalised the stress is. Second, Fusion-IEMOCAP appears only as an upper-bound reference and is excluded from the replication claim, since it was trained on the distribution it is evaluated on.

TABLE II
IEMOCAP replication ($n = 2{,}132$): predicted-versus-true Pearson r. For every out-of-domain checkpoint, valence ≫ arousal, and the bootstrap 95% CI on $r_v - r_a$ excludes zero.

Checkpoint	$\mathbf{r_v}$	$\mathbf{r_a}$	$\mathbf{r_v - r_a}$ (95% CI)
Fusion-MELD (shipped)	0.688	0.485	+0.202 [0.159, 0.244]
Fusion-Acted	0.662	0.326	+0.336 [0.293, 0.381]
Fusion-V2	0.674	0.404	+0.270 [0.225, 0.314]
Fusion-Combined	0.726	0.621	+0.104 [0.072, 0.140]
Fusion-IEMOCAP†	0.742	0.659	in-domain reference

†In-domain upper bound; excluded from the replication claim.

D. The split is encoder-specific

Is the valence edge from emotion pretraining, or just from scale? We answer with a matched-size ridge probe on IEMOCAP embeddings, comparing emotion2vec and WavLM at equal width (Table III, Fig. 5). At 1024 dimensions emotion2vec carries valence far better than WavLM-large (CCC 0.660 vs. 0.387), and the same holds against WavLM-base+—so the edge is not a size effect. Crucially the direction of the asymmetry is encoder-specific: emotion2vec favours valence, whereas WavLM favours arousal (0.604 vs. 0.344 at base size). The valence-over-arousal split is thus not a universal law of frozen speech encoders but a specific, useful property of emotion pretraining—which is exactly why valence-primary scoring is right for emotion2vec and would be wrong for WavLM.

E. The fix: valence-primary scoring

Acting on the diagnosis, we compare valence-primary against legacy arousal-based scoring by the stressed/calm

TABLE III
Matched-size ridge probe on IEMOCAP (CCC). emotion2vec carries valence far better; WavLM instead favours arousal.

Encoder	dim	valence CCC	arousal CCC
emotion2vec-plus-large	1024	0.660	0.570
WavLM-base+	768	0.344	0.604
WavLM-large	1024	0.387	0.580

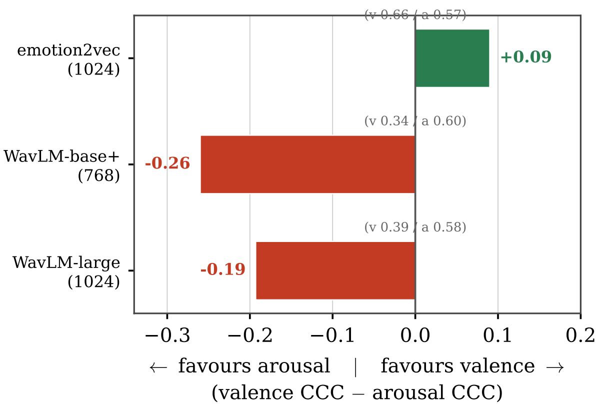

Fig. 5. Encoder-specific asymmetry. emotion2vec carries valence best; both WavLM variants carry arousal better than valence. The split is a property of emotion pretraining, not size or architecture.

separation d' over all six checkpoints (Fig. 6). Valence-primary scoring improves separation on every checkpoint, with a mean gain of +0.50 and a large effect (Cohen's $d = 1.71$; Wilcoxon signed-rank $p = 0.0312$, near the $n = 6$ floor of 0.0156 and reported for completeness, not as the main evidence). The gain is consistent in sign and large in size—the right claim for six paired observations.

F. Deployment accuracy

Under leave-one-out thresholding the shipped Fusion-MELD checkpoint reaches 91.7% on the real-voice set (Table IV). By language, English separates without error in this sample (1.000, bootstrap 95% CI $[1.000, 1.000]$) and Sinhala reaches 0.909 ($[0.727, 1.000]$). The wide Sinhala interval follows directly from small sample size and roughly three speakers, so we do not over-read the point estimate: English is strong and Sinhala is promising but under-determined, consistent with Sinhala being out of distribution for the frozen encoder.

V. Discussion

The five results tell one story. A frozen emotion encoder carries the valence of real stress reliably but its arousal only weakly, and that asymmetry is what turns in-domain acted CCC—which rewards arousal agreement on expressive speech—into an inverted guide for real deployment. Selecting on it favours checkpoints tuned to the acted high-arousal manifold, and those generalise worst to quiet real stress; the model trained on naturalistic conversation, weak in-domain, generalises best. Once stress is read from

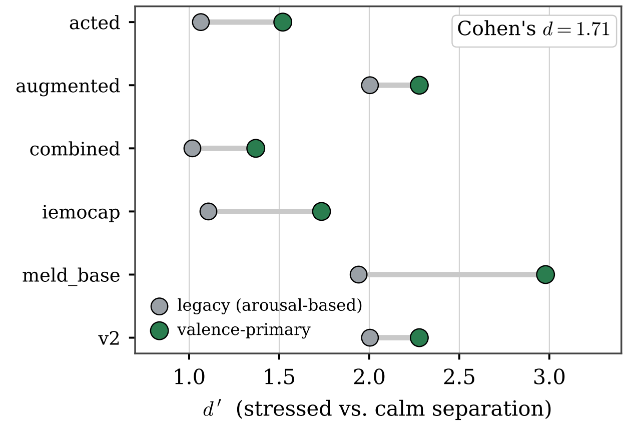

Fig. 6. Valence-primary scoring improves stressed/calm separation (d') over legacy arousal-based scoring for all six checkpoints (Cohen's $d = 1.71$).

TABLE IV
Deployment accuracy of the shipped model ($n = 24$). Overall is leave-one-out; per-language values are fixed-threshold with bootstrap 95% CIs.

Split	n	Accuracy	95% CI
All (LOO)	24	0.917	$[0.875, 1.000]$
English	13	1.000	$[1.000, 1.000]$
Sinhala	11	0.909	$[0.727, 1.000]$

valence, every checkpoint improves and the ordering stops hurting.

Why does valence survive the shift and arousal not? Arousal in speech rides mostly on prosodic energy and pitch dynamics—the very cues an acted delivery exaggerates and a quiet, internalised speaker suppresses; valence rides on more distribution-stable tonal and affective qualities that emotion pretraining captures well. The matched-size probe sharpens the point: the asymmetry belongs to emotion pretraining specifically, not to frozen encoders in general, since WavLM—pretrained for full-stack speech—shows the opposite bias. The lesson is not "always score on valence" but "know which axis your encoder transfers, and score on that."

The same logic sets bilingual expectations. English is well covered and separates cleanly; Sinhala is out of distribution, and no scoring rule changes the fact that the frozen encoder never saw it in pretraining. Rather than overclaim coverage, the system routes around the limit: voice supplies the valence estimate it can support, and arousal magnitude comes from the heart-rate-variability module. That is the right division of labour given the diagnosis, and it is why the paper is a design as much as a finding.

VI. Limitations and Future Work

The real-voice set is small ($n = 24$, roughly three Sinhala speakers). Leave-one-out thresholding removes fit-to-test bias but not small-sample variance, which is why the per-language intervals are wide and why we lean on the IEMOCAP replication for inference. Sinhala is out of distribution for the frozen encoder, and this cannot simply be trained away: adding 26 Sinhala clips to training lowered held-out accuracy from 90.9% to 81.8%—a controlled negative result showing that more in-language data cannot teach a frozen out-of-distribution encoder what its pretraining never encoded. In a small three-speaker probe we also saw a confident single-speaker misread—a quiet accented-English calm clip scored stressed (9.13 at confidence 0.91); the deployed product mitigates such cases with a faint-recording confidence guard, a speaker-relative (pre-to-post) verdict, and deferral to the physiological module, though the exact faint-audio threshold still awaits calibration on live-test data. Arousal magnitude is unreliable by design and is offloaded rather than fixed. Finally, Fusion-IEMOCAP is a reference upper bound, not part of the replication claim. Future work should grow the Sinhala speaker pool, calibrate the quality gate on live sessions, and test whether a lightly adapted (rather than fully frozen) encoder can recover arousal for internalised stress without losing the valence transfer that carries the system today.

VII. Conclusion

On real, spontaneous stress, ranking frozen speech-emotion models by their in-domain accuracy on acted speech is not just uninformative—it is inverted. The reason is a valence–arousal split: these embeddings carry the valence of real stress but not its arousal, a pattern we replicate on a large independent corpus and trace specifically to emotion pretraining. Reading stress from valence rather than arousal follows directly from the diagnosis and widens stressed/calm separation on every model we tested, giving a 91.7% leave-one-out detector deployed bilingually for English and Sinhala, with arousal deferred by design to a physiological channel. We recommend that future voice-stress systems on frozen emotion encoders first characterise which affective axis actually transfers to their deployment distribution, and select and score accordingly, rather than trusting in-domain acted agreement.

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